

# SecureAzCloud SCADA/ICS PQC Migration Continuity Checklist

Safety-first public checklist covering latency, legacy gateways, maintenance windows, vendor dependencies, hybrid mode, rollback, and safety continuity for OT/ICS/SCADA environments.

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This artifact is a public/redacted planning aid. It uses example rows and does not include any organization-specific secrets, hostnames, IP addresses, customer data, proprietary architecture, or key material.

## Purpose and Outcome

The SCADA/ICS PQC Migration Continuity Checklist converts PQC migration concepts into safety-first OT implementation controls. It focuses on latency, legacy gateways, maintenance windows, vendor dependencies, hybrid mode, rollback, lab validation, compensating controls, and safety continuity.

## Safety-First Migration Principles

- Do not test cryptographic changes first in production control environments.
- Define engineering-approved latency, jitter, timeout, CPU, memory, and bandwidth thresholds before testing.
- Require operations and safety signoff for maintenance windows, rollback plans, and stop conditions.
- Treat unsupported legacy gateways, protocol converters, firmware, and vendor tunnels as managed exceptions with compensating controls.
- Preserve operational continuity: availability and safety requirements can outweigh cryptographic modernization sequencing.

## Continuity Checklist Coverage

| Domain                  | Checklist Focus                                                                                                           | Evidence                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Governance              | Joint IT/OT ownership, safety-stop criteria, production change rules, rollback authority.                                 | Approved charter, change policy, safety review.          |
| Discovery               | Remote access, engineering workstations, HMIs, historians, gateways, PLC/RTU management, update channels, vendor tunnels. | OT crypto inventory and protocol map.                    |
| Latency and performance | Baseline latency, jitter, timeout, CPU, memory, bandwidth, packet loss, and allowed thresholds.                           | Baseline report and engineering-approved thresholds.     |
| Legacy gateways         | Protocol converters, serial-to-IP, VPN concentrators, firewall appliances, terminal servers, embedded crypto.             | Firmware inventory, support lifecycle, replacement plan. |
| Vendor dependencies     | PQC roadmap, cryptographic disclosure, support windows, firmware compatibility, rollback guidance.                        | Vendor response and support record.                      |
| Maintenance windows     | Production cycles, safety permits, staffing, vendor availability, downtime, rollback duration.                            | Change record and approved window.                       |
| Hybrid mode             | Non-production testing of hybrid classical/PQC where supported.                                                           | Lab result, compatibility matrix.                        |
| Rollback and continuity | Trust-store rollback, gateway config restore, cert rollback, firmware rollback, monitoring restore.                       | Rollback drill evidence.                                 |
| Compensating controls   | Segmentation, jump hosts, allowlisting, monitoring, vendor access control, encryption overlays, exception tracking.       | Exception register and control evidence.                 |

## Representative Test Cases

| Test                                                                | Metric                                                   | Acceptance Criteria                                                                    |
|---------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|
| TLS/mTLS gateway handshake under representative maintenance traffic | Handshake success, latency, CPU, memory, log visibility  | No critical errors; latency and resource usage within engineering-approved thresholds. |
| Certificate chain-size and trust-store compatibility                | Validation success, error rate, event logs               | All test clients validate or documented exception exists; errors are observable.       |
| Legacy gateway rollback                                             | Rollback time, configuration integrity, restored service | Rollback completes within approved maintenance window and restores previous behavior.  |
| Vendor remote access through controlled path                        | Connectivity, latency, MFA/allowlist status, monitoring  | No bypass of remote access controls; sessions logged and revocable.                    |
| Hybrid mode trial where vendor-supported                            | Protocol success, latency, error rate, fallback behavior | Hybrid mode works in lab or is blocked with evidence and compensating control plan.    |

## Deliverables in the XLSX Template

- Continuity\_Checklist: safety-first migration control list with status, priority, maintenance window, owner, and evidence.
- OT\_Crypto\_Inventory: OT/ICS asset and protocol inventory with latency tolerance, legacy gateway flag, vendor roadmap, hybrid mode, rollback, and compensating controls.
- Latency\_Test\_Cases: test scenarios and acceptance criteria for handshake, certificate, gateway, remote access, monitoring, and hybrid-mode trials.
- Vendor\_Roadmap: product/firmware support window, PQC roadmap, cryptographic disclosure, maintenance-window and rollback guidance.
- Sources: NIST, CISA, OT, and supply-chain reference map.

## Source and Standards Map

Public sources used to align this artifact. URLs are included for auditability.

| Source                                                                                  | Artifact Alignment                                                                                                               | Public URL                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NIST FIPS 203 - Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM)      | PQC key establishment/key encapsulation planning for TLS, VPN, API, and machine-to-machine trust dependencies.                   | <a href="https://csrc.nist.gov/pubs/fips/203/final">https://csrc.nist.gov/pubs/fips/203/final</a>                                                                                                                       |
| NIST FIPS 204 - Module-Lattice-Based Digital Signature Standard (ML-DSA)                | Digital signature migration planning for certificates, code signing, artifact signing, identity token signing, and PKI roadmaps. | <a href="https://csrc.nist.gov/pubs/fips/204/final">https://csrc.nist.gov/pubs/fips/204/final</a>                                                                                                                       |
| NIST FIPS 205 - Stateless Hash-Based Digital Signature Standard (SLH-DSA)               | Hash-based signature planning for conservative signature-use cases where size/performance tradeoffs are acceptable.              | <a href="https://csrc.nist.gov/pubs/fips/205/final">https://csrc.nist.gov/pubs/fips/205/final</a>                                                                                                                       |
| NIST NCCoE Migration to Post-Quantum Cryptography                                       | Cryptographic discovery, inventory, risk-prioritization, and interoperability-testing workflow.                                  | <a href="https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc">https://www.nccoe.nist.gov/applied-cryptography/migration-to-pqc</a>                                                                         |
| NIST CSWP 39 - Considerations for Achieving Crypto Agility                              | Crypto-agility strategy, governance, algorithm/configuration abstraction, operational continuity, and lifecycle planning.        | <a href="https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.39.pdf">https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.39.pdf</a>                                                                                           |
| CISA - Product Categories for Technologies That Use Post-Quantum Cryptography Standards | 2026 product-category reference for PQC adoption, including cloud services, web software, and endpoint security categories.      | <a href="https://content.govdelivery.com/accounts/USDHSCISA/bulletins/405c246">https://content.govdelivery.com/accounts/USDHSCISA/bulletins/405c246</a>                                                                 |
| CISA/NSA/NIST - Quantum-Readiness: Migration to Post-Quantum Cryptography               | Quantum-readiness roadmap, cryptographic inventory, vendor engagement, and supply-chain dependency assessment.                   | <a href="https://www.cisa.gov/resources-tools/resources/quantum-readiness-migration-post-quantum-cryptography">https://www.cisa.gov/resources-tools/resources/quantum-readiness-migration-post-quantum-cryptography</a> |
| CISA - Post-Quantum Considerations for Operational Technology                           | OT/ICS/SCADA migration considerations for safety, performance, vendor dependency, and operational continuity.                    | <a href="https://www.cisa.gov/resources-tools/resources/post-quantum-considerations-operational-technology">https://www.cisa.gov/resources-tools/resources/post-quantum-considerations-operational-technology</a>       |
| NIST SP 800-82 Rev. 3 - Guide to Operational Technology Security                        | OT security planning that accounts for performance, reliability, safety, and physical process constraints.                       | <a href="https://csrc.nist.gov/pubs/sp/800/82/r3/final">https://csrc.nist.gov/pubs/sp/800/82/r3/final</a>                                                                                                               |
| NIST SP 800-161 Rev. 1 Update 1 - Cybersecurity Supply Chain Risk Management Practices  | Supplier roadmap, procurement, support-window tracking, and supply-chain risk-management evidence.                               | <a href="https://csrc.nist.gov/publications/search?keywords=lg=800-161">https://csrc.nist.gov/publications/search?keywords=lg=800-161</a>                                                                               |
| NIST Cybersecurity Framework 2.0                                                        | Govern, Identify, Protect, Detect, Respond, and Recover framing for migration governance and risk communication.                 | <a href="https://www.nist.gov/cyberframework">https://www.nist.gov/cyberframework</a>                                                                                                                                   |